

CNAM Paris - Projet NFE204

Introduction à l'algorithme k-means avec Apache Mahout

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21 janvier 2014



Introduction à l'algorithme k-means avec Apache Mahout

- Introduction à Apache Mahout
- Clustering (concepts & applications)
- Introduction à l'algorithme k-means
- Apache Mahout k-means
- k-means et MapReduce
- k-means – Processus E2E
 - Exemple fichiers Reuters

Introduction à Apache Mahout

- Projet de la Fondation Apache.
- Bibliothèque **open source** d'apprentissage automatique
- Framework d'outils pour implémentation d'algorithmes de **Machine Learning** en Java
- La plupart implémentée avec Apache Hadoop en utilisant le paradigme MapReduce
- Encore récent, mais en croissance rapide (depuis début 2009)



Introduction à Apache Mahout

- Familles des principaux algorithmes
 - *Classification*
 - *Recommenders / Collaborative Filtering*
 - *Clustering*



Introduction à Apache Mahout

Classification

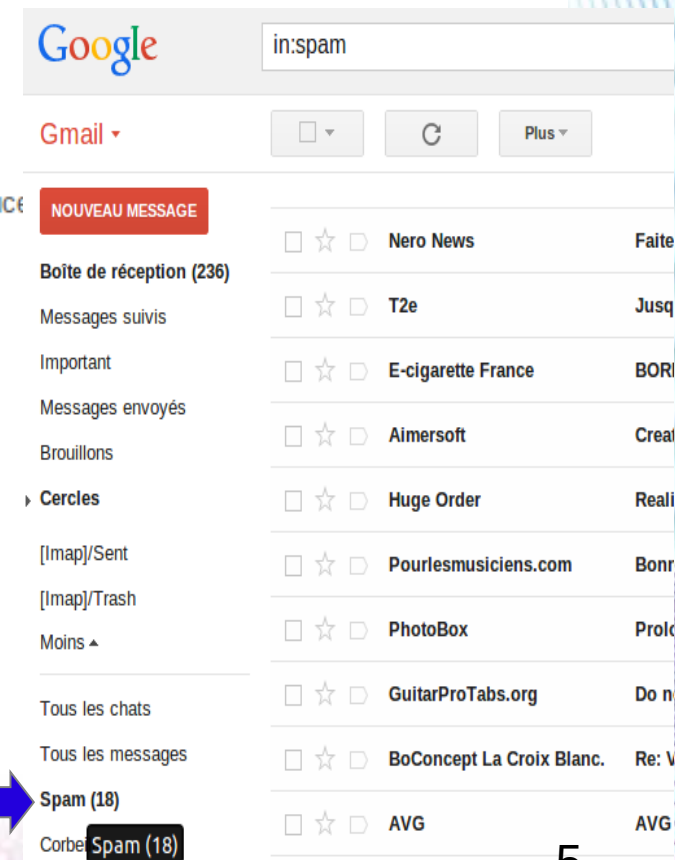
A general introduction to the most common text classification algorithms can be found at Google Answers: <http://answers.google.com/answers/main?cmd=threadview&id=225316> For information on the algorithms implemented in Mahout (or scheduled for implementation) please visit the following pages.

Fully supported:

- Logistic Regression (SGD) - model parameter selection can be done in Hadoop
- Naive Bayes/ Complementary Naive Bayes - training runs on Hadoop
- Random Forests (MAHOUT-122, - training is done in Hadoop MAHOUT-140, MAHOUT-145)
- Hidden Markov Models (see MAHOUT-627, MAHOUT-396, MAHOUT-734) - training is done in Map-Reduce

Deprecated or drafts only:

- Support Vector Machines (see MAHOUT-14 , MAHOUT-232 and MAHOUT-334)
- Perceptron and Winnow (see MAHOUT-85)
- Neural Network (see MAHOUT-228)
- Restricted Boltzmann Machines (see MAHOUT-375)
- Online Passive Aggressive (see MAHOUT-702)
- Boosting (see MAHOUT-716)



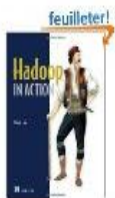
Introduction à Apache Mahout

Recommenders / Collaborative Filtering

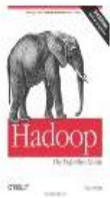
Mahout contains both simple non-distributed recommender implementations and distributed Hadoop-based recommenders.

- First-timer FAQ
- Non-distributed recommenders ("Taste")
- Distributed Item-Based Collaborative Filtering
- Collaborative Filtering using a parallel matrix factorization

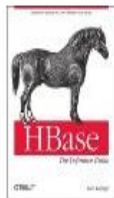
Les clients ayant acheté cet article ont également acheté



Hadoop in Action
› Chuck Lam
★★★★☆ (1)



Hadoop - The Definitive Guide 3e
› Tom White



HBase: The Definitive Guide
› Lars George
Broché



Taming Text How to Find, Organize and ...
› Grant S Ingersoll



Programming Hive
Edward Rutherglen
Broché



Machine Learning in Action
› Peter Harrington
Broché



Programming Pig
› Alan Gates
Broché



Activiti in Action
Mr Tijs Rademakers
Broché

Introduction à Apache Mahout

Clustering

For a more detailed explanation see [Wikipedia page](#) or checkout our [Reference Reading](#)

Fully supported:

- MAHOUT:Canopy Clustering (MAHOUT-3 - runs on Hadoop)
- K-Means Clustering (MAHOUT-5 - runs on Hadoop)
- Fuzzy K-Means (MAHOUT-74 - runs on Hadoop)
- [Expectation Maximization](../clustering/expectation-maximization.html) (MAHOUT-28 - runs on Hadoop)
- Mean Shift Clustering (MAHOUT-15 - runs on Hadoop)
- Dirichlet Process Clustering (MAHOUT-30 - runs on Hadoop)
- Latent Dirichlet Allocation (MAHOUT-123) - runs on Hadoop
- Minhash Clustering (MAHOUT-344) - runs on Hadoop
- kMeans++ streaming clustering - documentation missing

Deprecated or drafts only:

- Hierarchical Clustering (MAHOUT-19, MAHOUT-843)
- Spectral Clustering (MAHOUT-363)

YAHOO! FRANCE

Rechercher

Connexion

Votre avis - Choisir sa put

Mail

Actualités

Sport

Pour Elles

Finance

Cinéma

TV

Musique

Auto

Screen

Vous l'ignorez, mais ils sont cancérogènes

De nombreux produits que vous pensez inoffensifs sont en réalité hautement cancérogènes.

Fuyez-les »

1 - 5 sur 50

Tendances du jour

1 Carla Bruni	6 Prêt en ligne
2 Frédéric Taddei	7 Victoire Ana Ivanovic
3 Téléviseur LED	8 Soldes chaussures
4 Affaire Zahia	9 Ligue 1
5 Michaël Blanc libre	10 Inondations Var

Les vidéos du jour Yahoo Screen

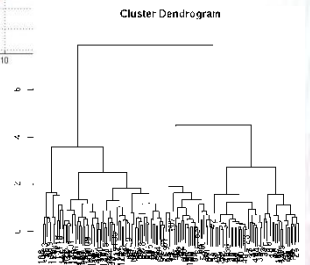
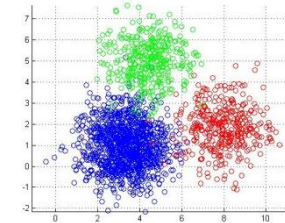
Clustering (concepts & applications)

- **Hard Clustering**

- Répartition en K classes disjointes d'une collection de documents

- **Partitionnement de documents**

- Partitionnement à plat
- Partitionnement hiérarchique



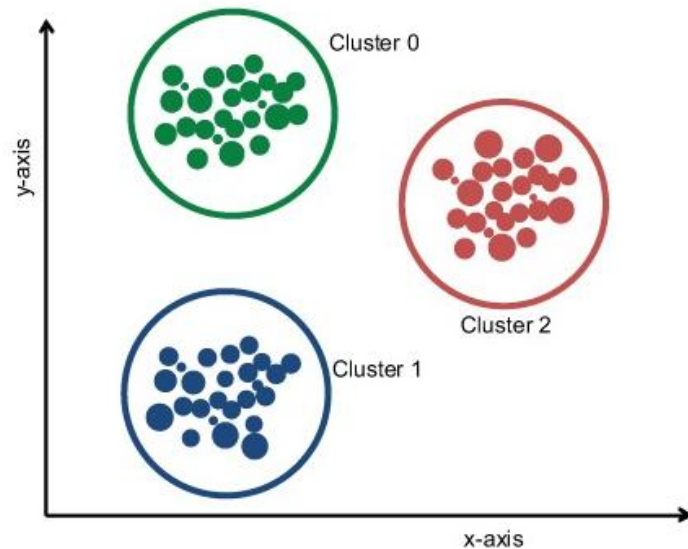
- **Classification non supervisée dure**

- Applicable à toute collection d'objets
 - textes, images, vidéos, logs

Clustering (concepts & applications)

- **Marketing:** segmentation du marché, positionnement du produit
- La **segmentation d'image:** traitement d'images pour rassembler des pixels entre eux suivant des critères pré-définis
- **Données hospitalières:** tendance service au patient, propagation d'épidémie
- *Scientific Imaging*, séquençage de l'ADN
- Analyses des **réseaux sociaux**

Introduction à l'algo k-means

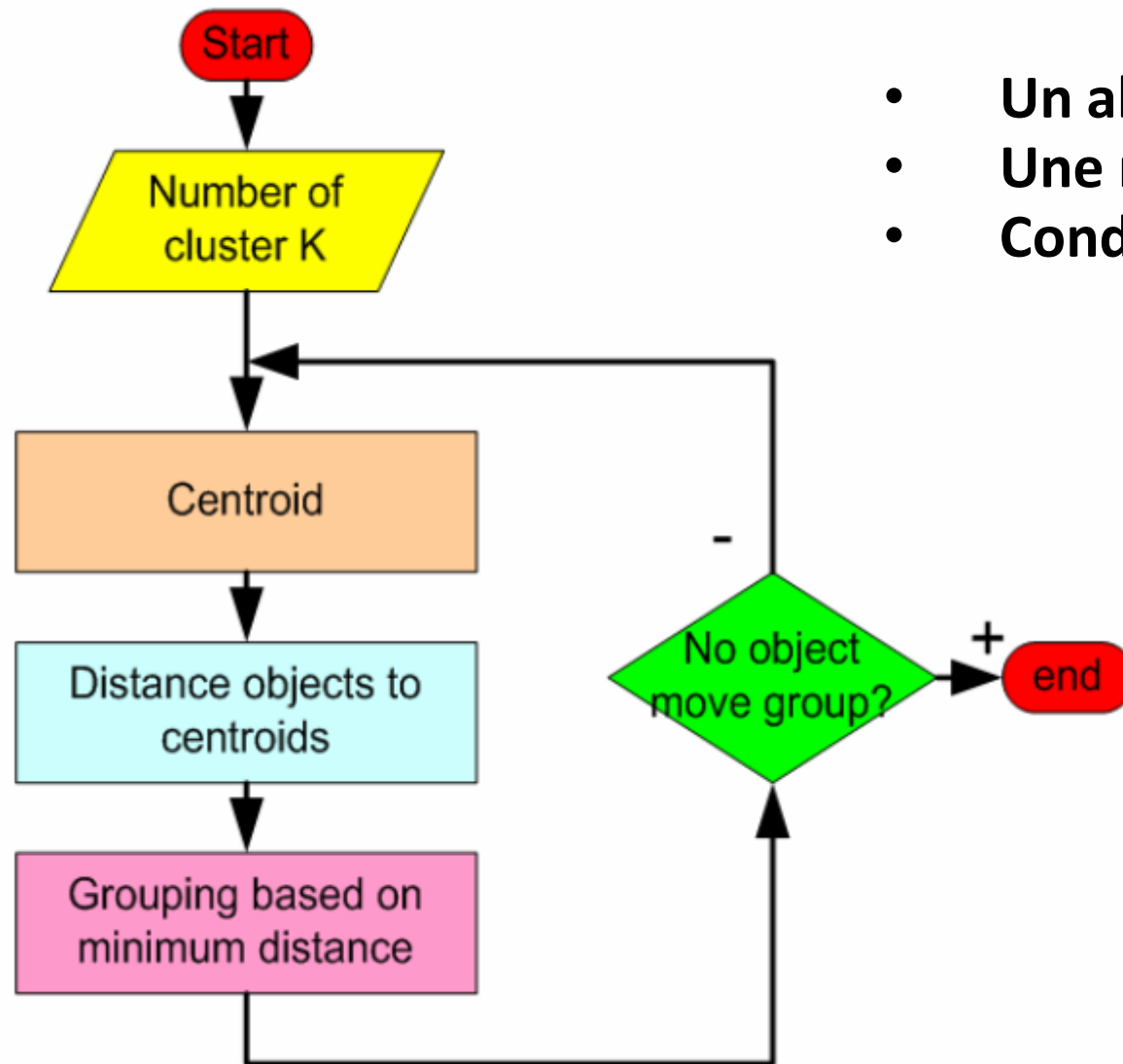


POUR
LES NULS[®]

Introduction à l'algo k-means

- Appartient à la famille **Clustering**
- **Partionnement à plat**: méthodes de réallocation; non hiérarchique
- Faire émerger des groupes homogènes (ayant des **caractéristiques communes**) des données collectées
- **Les nuées dynamiques** sont une généralisation de ce principe

Introduction à l'algo k-means



- Un algorithmme
- Une notion de similarité
- Condition d'arrêt

Introduction à l'algo k-means

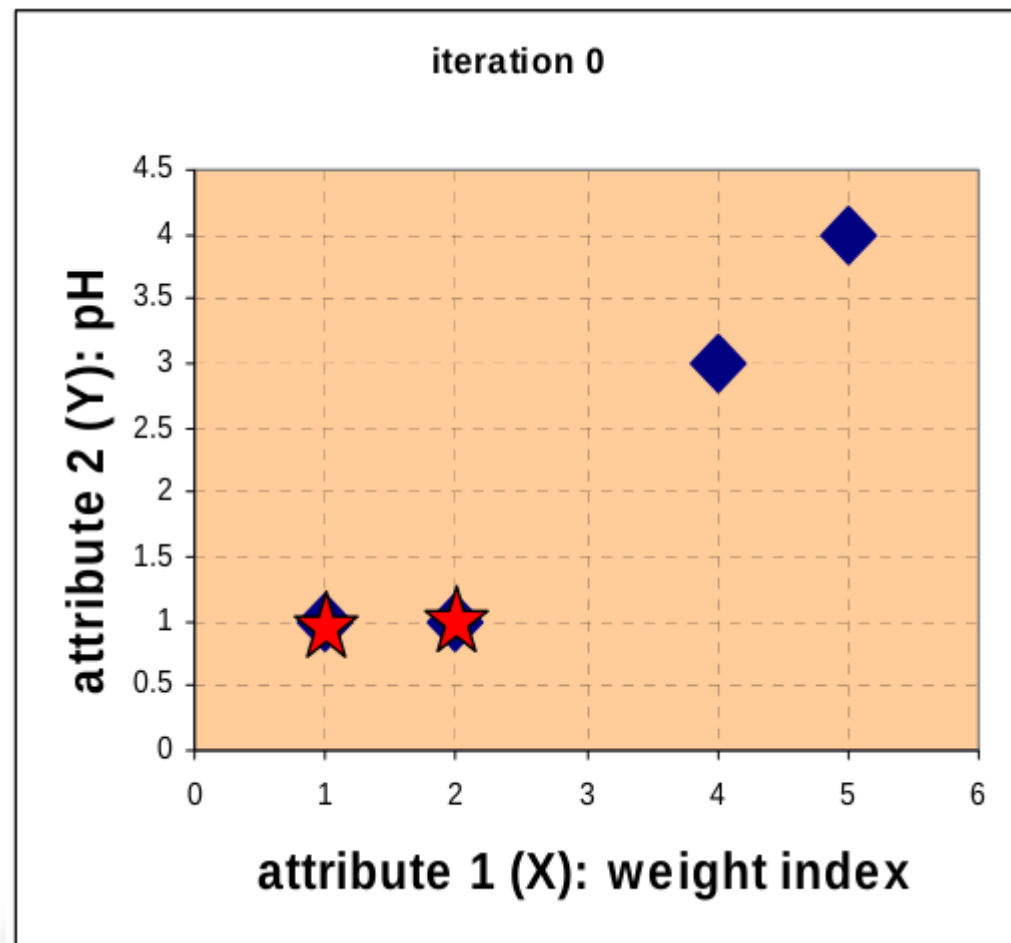
Object	Feature 1 (X): weight index	Feature 2 (Y): pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

A	B	C	D	
1	2	4	5	X
1	1	3	4	Y
◆	◆	◆	◆	

K = 2

Valeurs initiales des centres :

c1 = (1,1) et c2 = (2,1)



Introduction à l'algo k-means

Object	Feature 1 (X): weight index	Feature 2 (Y): pH
Medicine A	1	1
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Medicine C	4	3
Medicine D	5	4

Distance objet-centre:

$$D^0 = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} c1 \\ c2 \end{matrix} & \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 1 & 0 & 2.83 & 4.24 \end{bmatrix} \end{matrix}$$

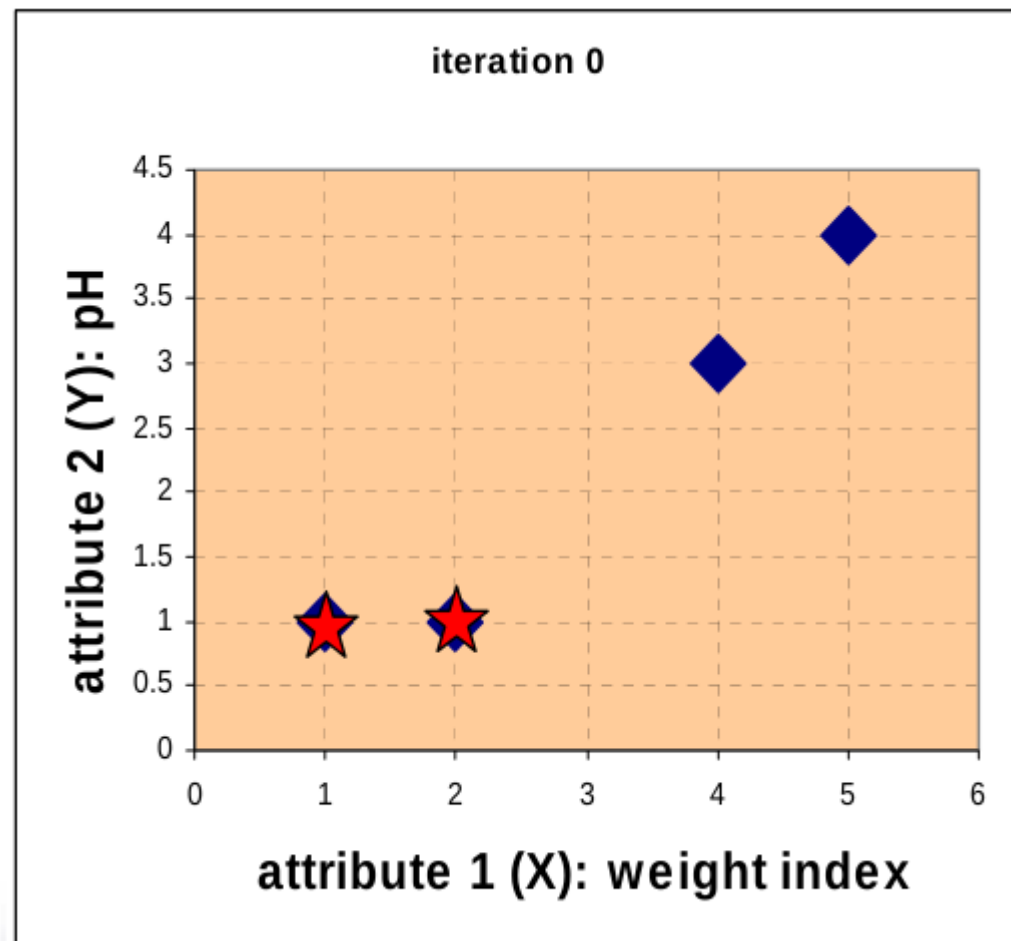
distance de C(4,3) à

$$c1 \rightarrow \sqrt{(4-1)^2 + (3-1)^2} = 3.61$$

$$c2 \rightarrow \sqrt{(4-2)^2 + (3-1)^2} = 2.83$$

Clustering:

$$G^0 = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} c1 \\ c2 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$



Introduction à l'algo k-means

Object	Feature 1 (X): weight index	Feature 2 (Y): pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
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Clustering:

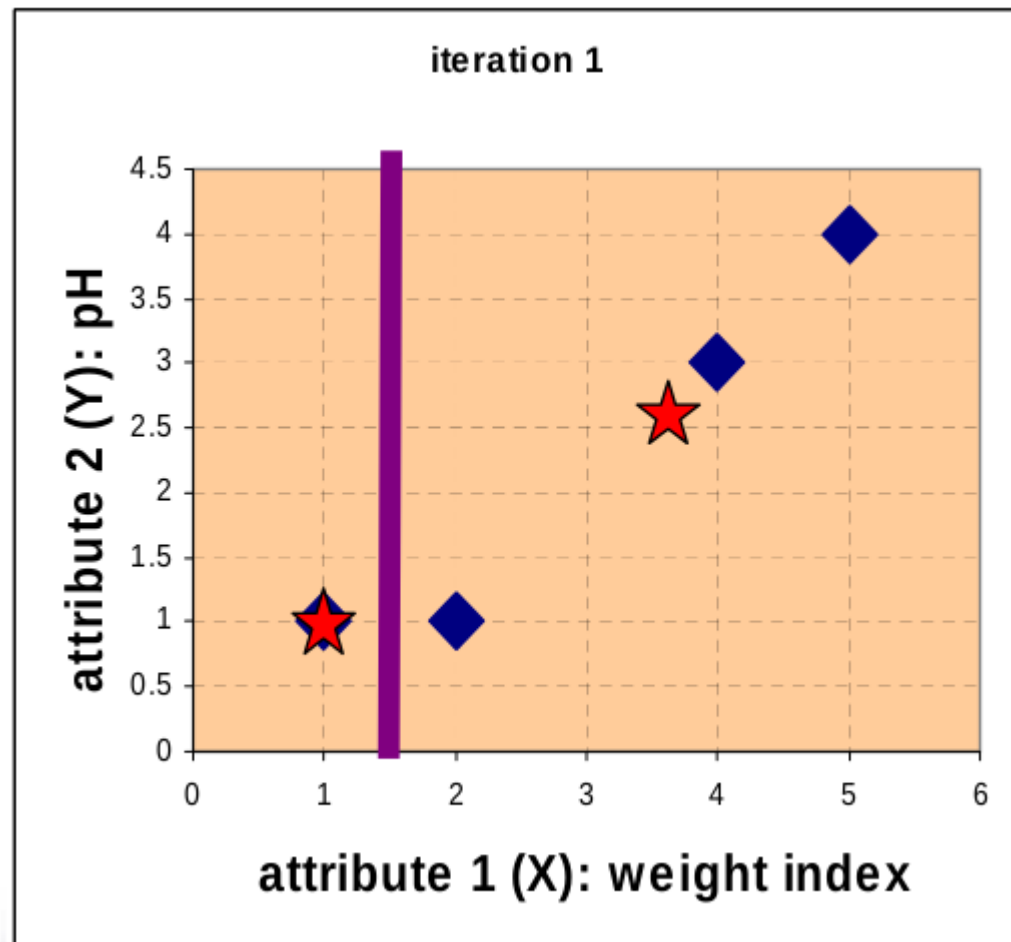
$$G^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

A B C D

Itération 1 - nouveaux centres:

$$c_1 = (1, 1) \quad \star$$

$$c_2 = \left(\frac{2+4+5}{3}, \frac{1+3+4}{3} \right) = \left(\frac{11}{3}, \frac{8}{3} \right) \quad \star$$



Introduction à l'algo k-means

Object	Feature 1 (X): weight index	Feature 2 (Y): pH
Medicine A	1	1
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Itération 1 - Distance objet-centre:

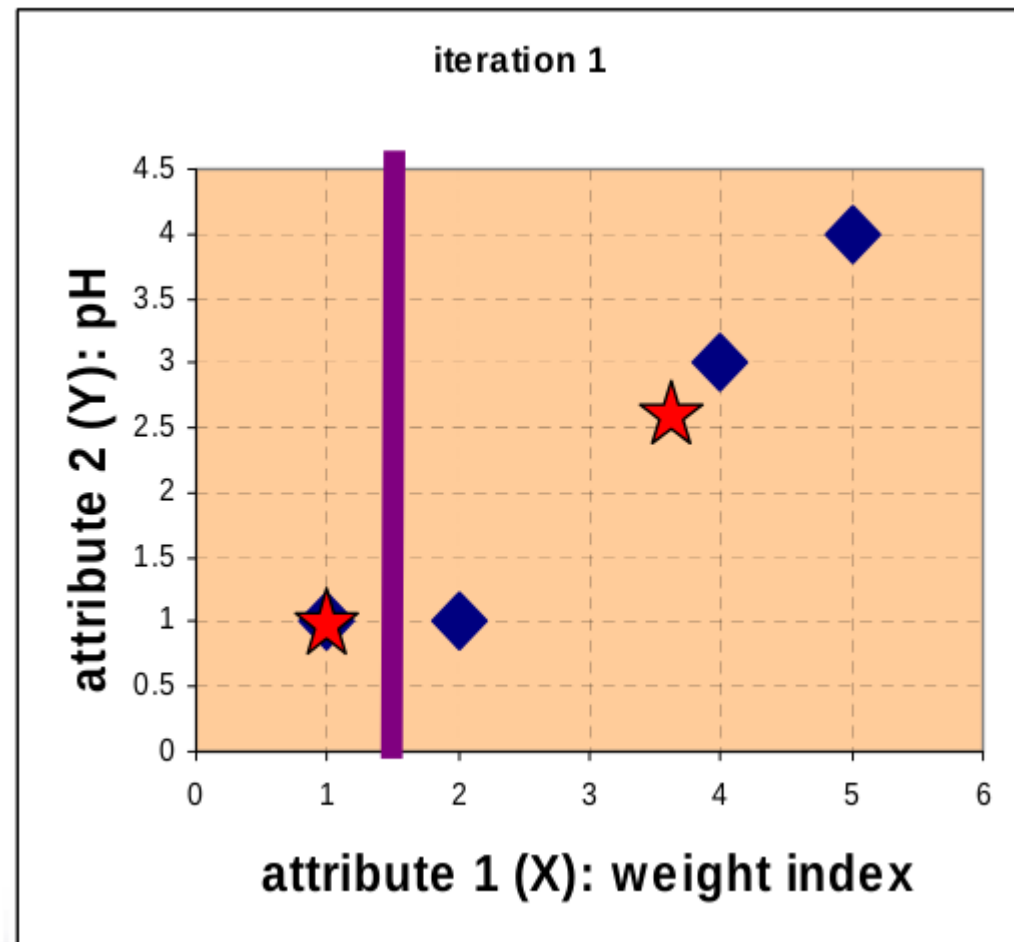
$$D^1 = \begin{bmatrix} A & B & C & D \\ 0 & 1 & 3.61 & 5 \\ 3.14 & 2.36 & 0.47 & 1.89 \end{bmatrix} \begin{matrix} c1 \\ c2 \end{matrix}$$

Itération 1 - Clustering:

$$G^1 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

A B C D

2 clusters



Introduction à l'algo k-means

Object	Feature 1 (X): weight index	Feature 2 (Y): pH
Medicine A	1	1
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Medicine D	5	4

Itération 1 - Clustering:

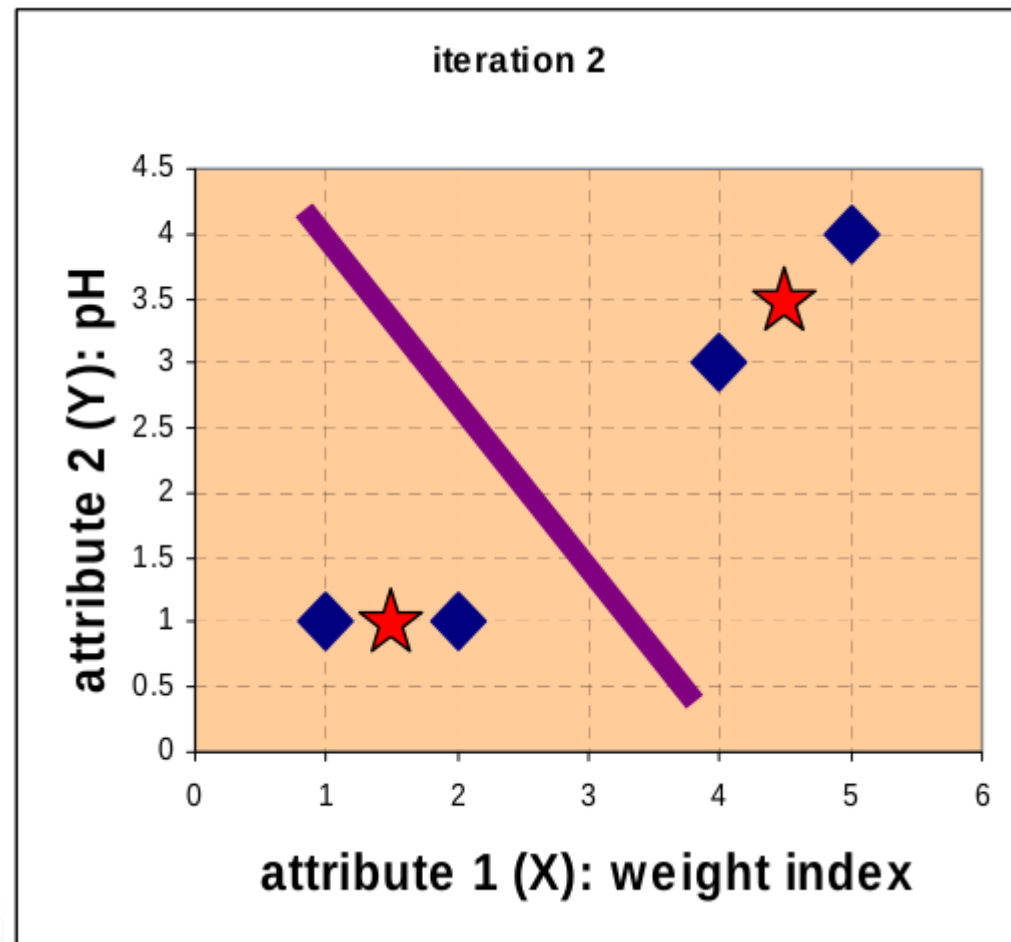
$$G^1 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

A B C D

Itération 2 - nouveaux centres:

$$c_1 = \left(\frac{1+2}{2}, \frac{1+1}{2} \right) = \left(1\frac{1}{2}, 1 \right) \quad \star$$

$$c_2 = \left(\frac{4+5}{2}, \frac{3+4}{2} \right) = \left(4\frac{1}{2}, 3\frac{1}{2} \right) \quad \star$$



Introduction à l'algo k-means

Object	Feature 1 (X): weight index	Feature 2 (Y): pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

Itération 2 - Distance objet-centre:

$$D^2 = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} c1 \\ c2 \end{matrix} & \begin{bmatrix} 0.5 & 0.5 & 3.20 & 4.61 \\ 4.30 & 3.54 & 0.71 & 0.71 \end{bmatrix} \end{matrix}$$

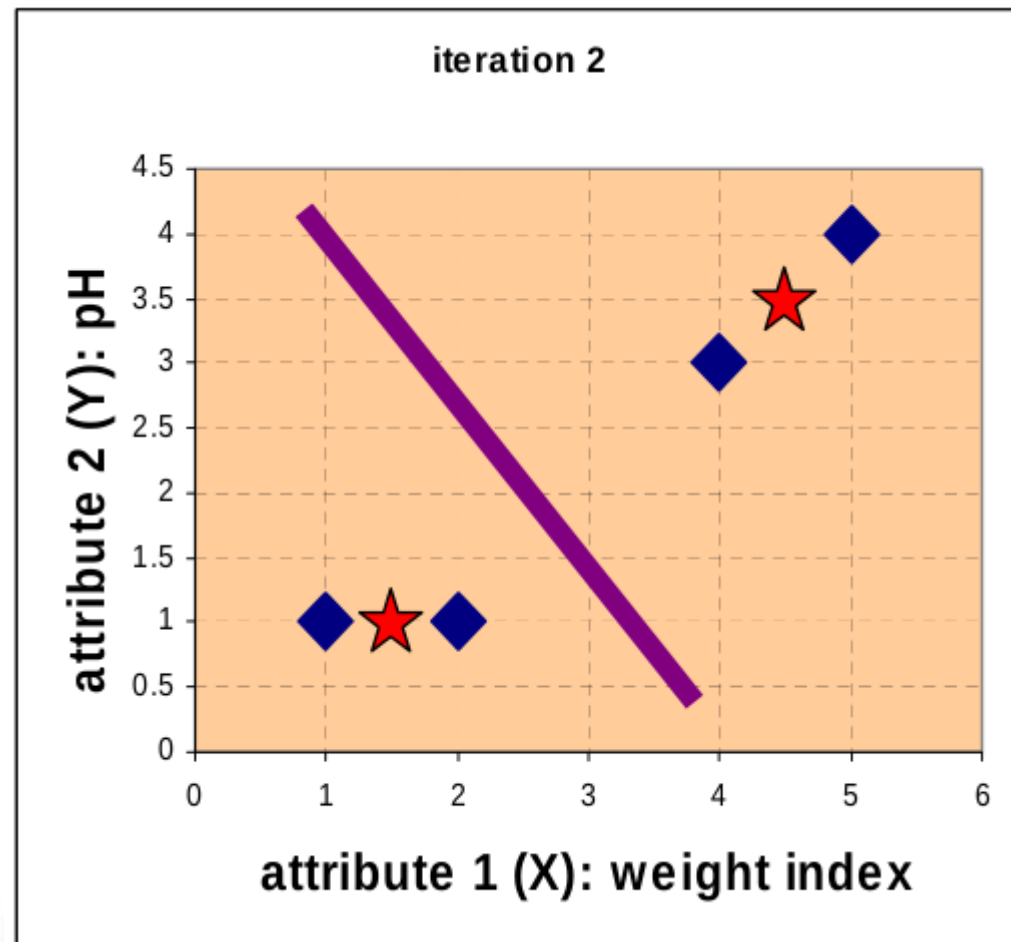
Itération 2 - Clustering:

$$G^2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

A B C D

2 clusters

$G^2 = G^1 \implies \text{STOP !}$



Introduction à l'algo k-means

Remarques:

- **Complexité importante**
 - $k \times n \times O(\text{distance}) \times \text{nb d'itérations}$
- **La solution finale dépend des représentants initiaux choisis;**
un choix aléatoire rend l'algorithme non déterministe
- **Pas de minimum global** de la distance intra-classe, seul le minimum local est atteint.

Apache Mahout k-means

Apache Mahout k-means

```
bin/mahout kmeans \
```

```
-i <input vectors directory> \  
-c <input clusters directory> \  
-o <output working directory> \
```

```
-k <optional number of initial clusters to sample from input vectors> \  
-dm <DistanceMeasure> \  
-x <maximum number of iterations> \
```

Apache Mahout k-means

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bin/mahout kmeans \
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-x <maximum number of iterations> \
```

- Le nombre de centres de gravité K
 - Peuvent être générés de manière aléatoire avec RandomSeedGenerator
- Quelle mesure de distance ?
 - *Euclidean, Squared Euclidean, Manhattan, Cosine, Tanimoto, Weighted*
- Quel est le maximum nombre d'itérations ?

Apache Mahout k-means

bin/mahout **kmeans** \

```
-i <input vectors directory> \  
-c <input clusters directory> \  
-o <output working directory> \
```

```
-k <optional number of initial clusters to sample from input vectors> \  
-dm <DistanceMeasure> \  
-x <maximum number of iterations> \
```

- Où se trouvent les vecteurs ?
- Où se trouvent les centres (initiaux) ?
- Où seront générés les clusters par itération ?

Apache Mahout k-means

```
$MAHOUT kmeans \
```

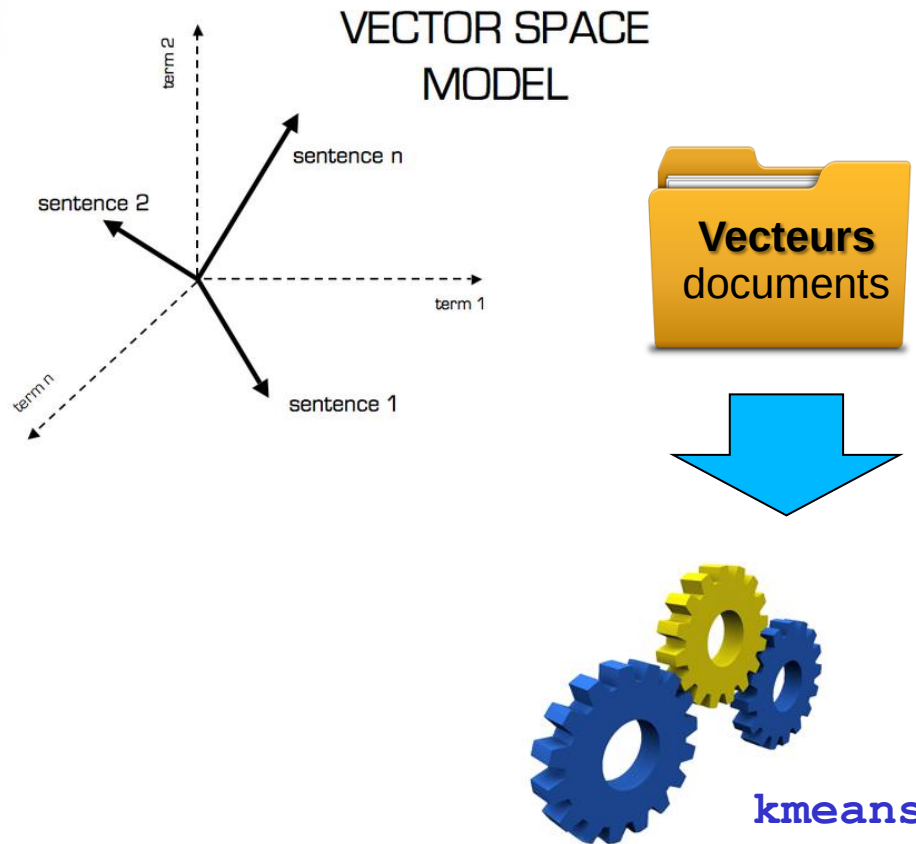
```
-i ${WORK_DIR}/reuters-out-seqdir-sparse-kmeans/tfidf-vectors/ \
-c ${WORK_DIR}/reuters-kmeans-clusters \
-o ${WORK_DIR}/reuters-kmeans \
```

```
-dm org.apache.mahout.common.distance.CosineDistanceMeasure \
-x 10 \
-k 20 \
```

```
-ow \
--clustering \
```

- **K = 20** centres de gravité
- Max nombre d'itérations = **10**
- Mesure de distance = **Cosinus**

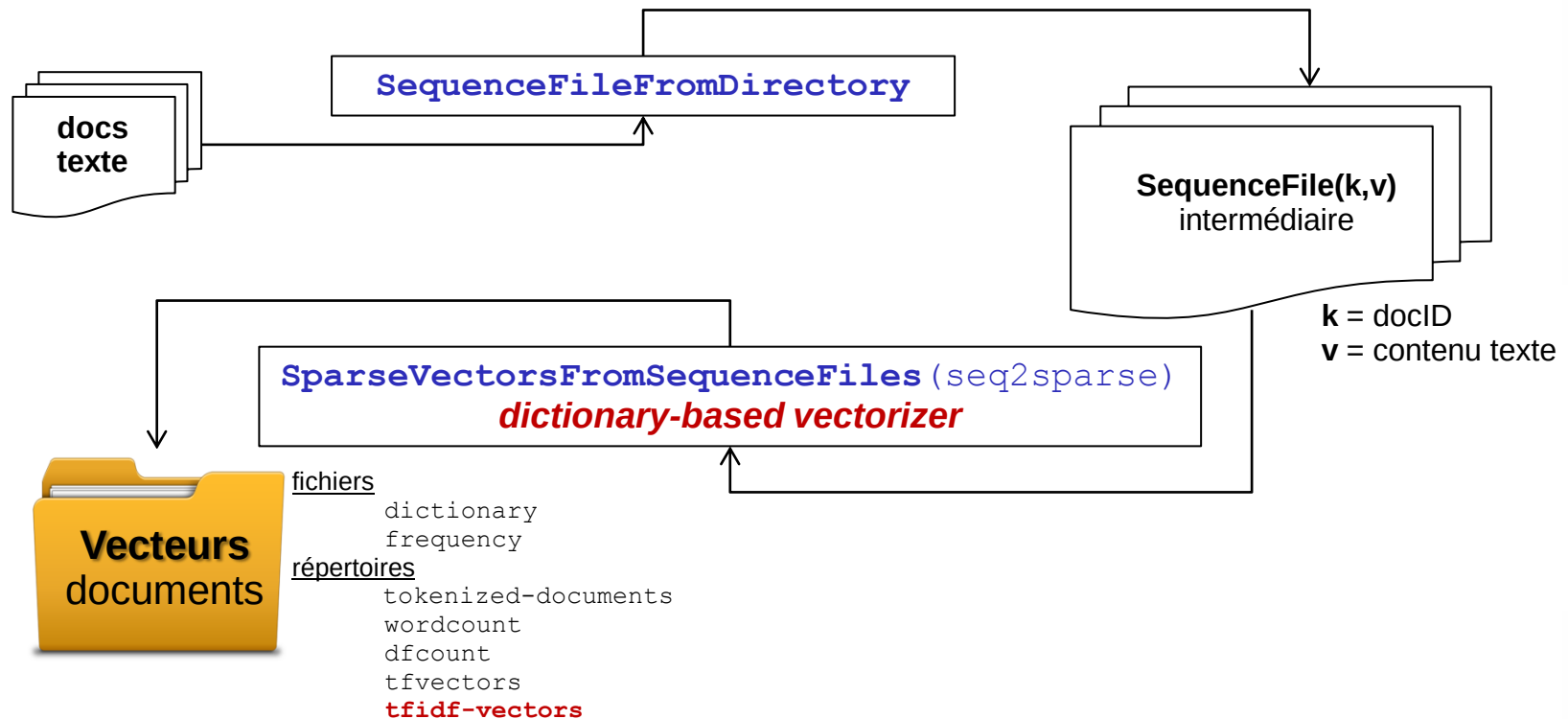
Apache Mahout k-means



id	men	entered	bank	charlotte	missiles	masks	aryan	guns	witnesses	reported	silve
seg1.txt	0.239441	0	0.153457	0.195243	0	0.237029	0	0.195243	0.237029	0.140004	0.1
seg13.txt	0	0	0	0	0	0	0	0	0	0	0
seg14.txt	0	0.192197	0	0	0	0	0	0	0	0	0
seg15.txt	0	0	0	0	0	0	0	0	0	0	0
seg16.txt	0	0	0	0	0	0	0	0	0	0	0
seg17.txt	0	0	0	0	0	0	0	0	0	0	0
seg18.txt	0	0.158432	0	0	0	0	0	0	0	0	0
seg19.txt	0	0	0	0.197255	0	0	0	0	0	0.141447	0
seg2.txt	0	0	0	0	0	0	0	0	0	0	0
seg20.txt	0	0.234323	0	0	0	0	0	0	0	0	0
seg21.txt	0	0	0	0	0	0	0	0	0	0	0
seg22.txt	0	0	0	0	0.139629	0	0.127389	0	0	0	0
seg23.txt	0	0	0	0	0	0	0	0	0	0.180656	0
seg24.txt	0	0	0	0	0	0	0.117966	0	0	0.117966	0

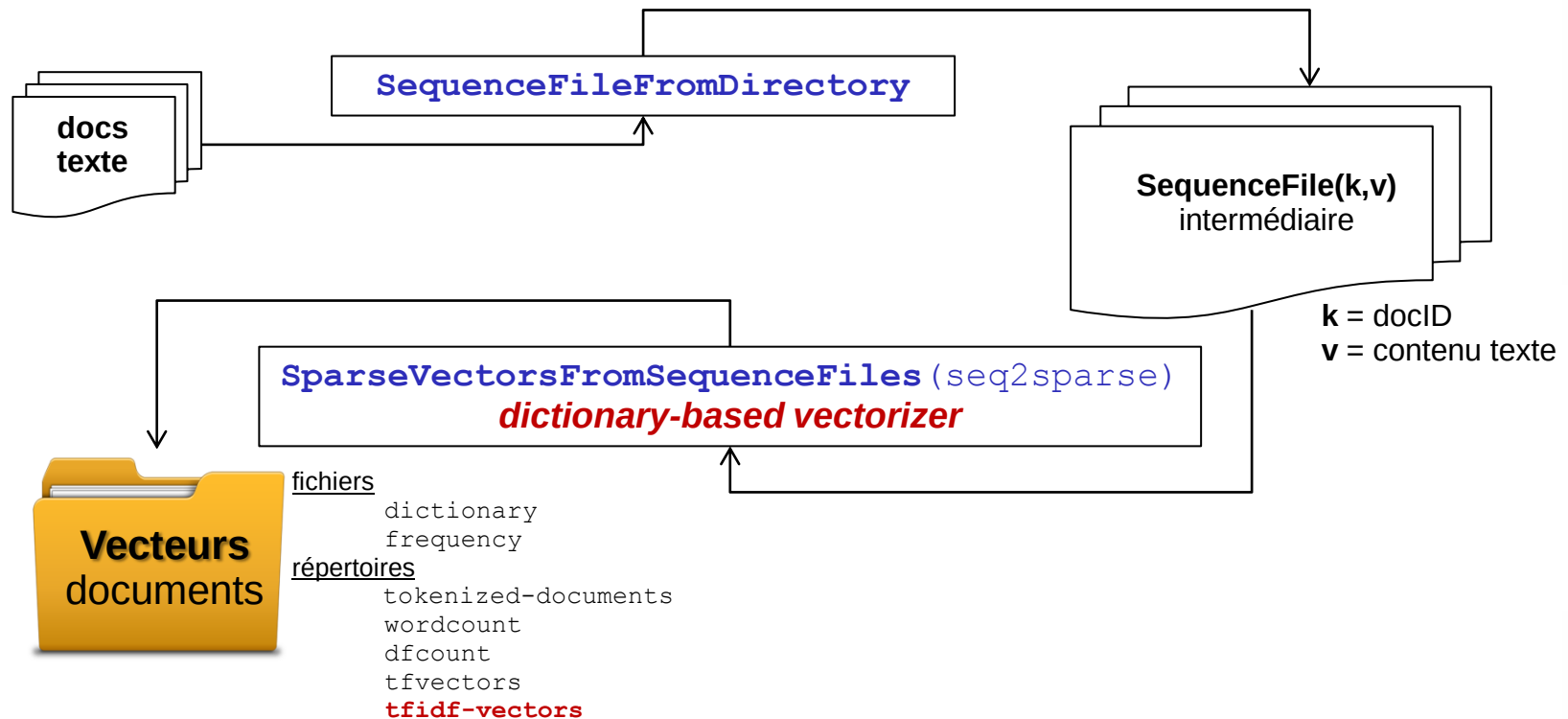
- Comment Mahout transforme-t-il les documents en vecteurs ?

Apache Mahout k-means



- Segmentation (*tokenization*)
- Comptage des mots + *n-gram collocations*
- Pondération **TF-IDF** par défaut

Apache Mahout k-means



Quelques paramètres pour **seq2sparse**

-a	tokenize words e.g. white space with <code>org.apache.lucene.analysis.WhitespaceAnalyzer</code>
-chunk	chunk size in MB
-wt	TF-IDF (default)
-s	Min support value
-md	Min doc freq value
-x	Max doc freq value
-ng	n-gram size
-ml	Min log-likelihood ratio (LLR)
-seq	SequentialAccessSparseVector flag
-norm	Normalisation

Lucene

k-means et MapReduce

k-means et MapReduce

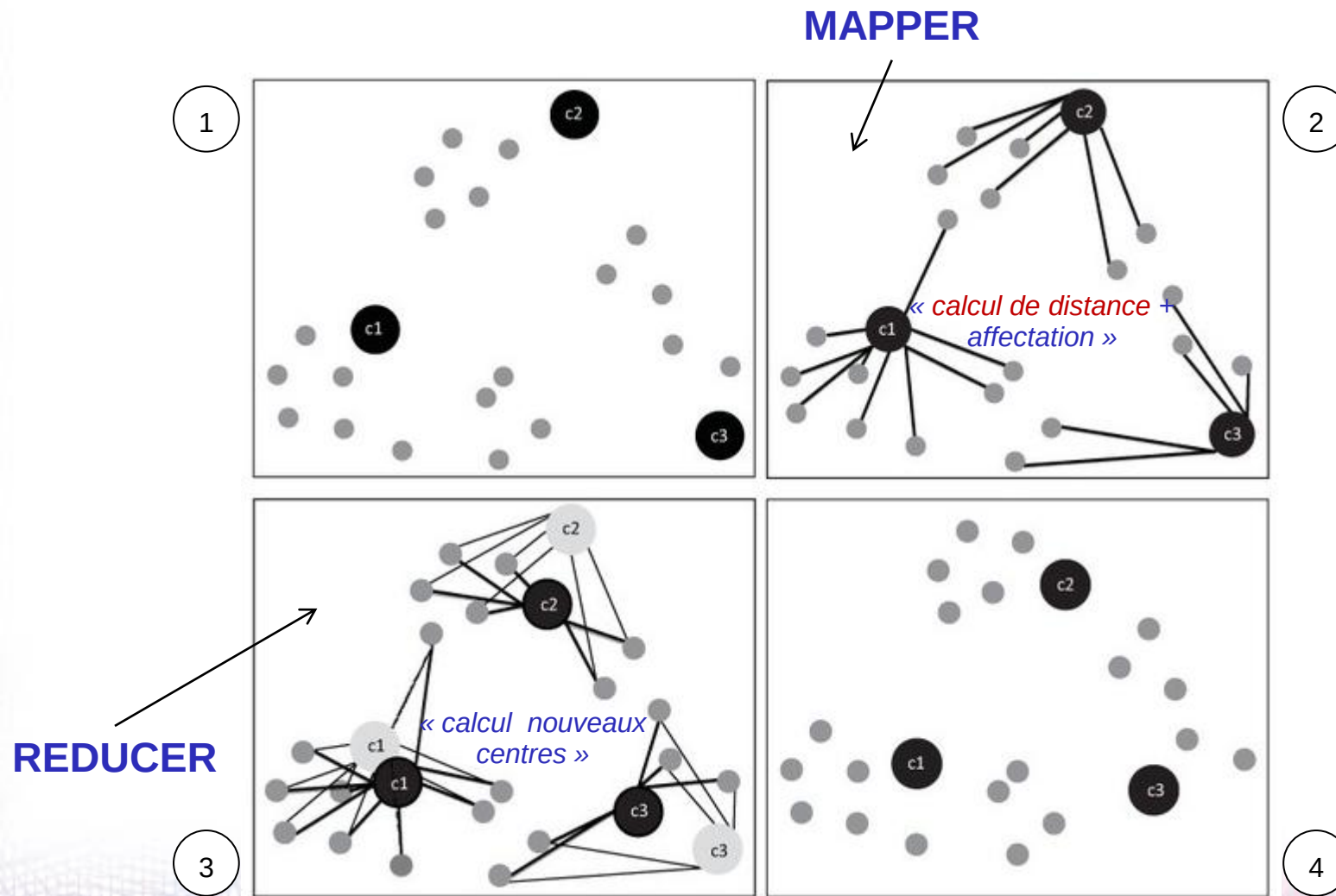
- Pour un ensemble de données dont taille $\sim T_o$, la **version MapReduce de k-means** offre une *scalabilité* en utilisant le Hadoop HDFS
- A chaque itération on a un total de **$n \times k$ calculs de distance**. n : nombre de points ou vecteurs
 - Le calcul de distance d'un vecteur et un centre étant indépendant de celui d'un autre vecteur et un centre $\rightarrow$ paralléliser.

k-means et MapReduce

- Pour un ensemble de données dont taille $\sim T_o$, la **version MapReduce de k-means** offre une *scalabilité* en utilisant le Hadoop HDFS
- A chaque itération on a un total de **$n \times k$ calculs de distance**. n : nombre de points ou vecteurs
 - Le calcul de distance d'un vecteur et un centre étant indépendant de celui d'un autre vecteur et un centre $\rightarrow$ **paralléliser**.
- k-means est un algorithme à 2 étapes:
 1. **Trouver les points proche d'un centre** et les affecter au cluster correspondant $\rightarrow$ **MAPPER**
 2. Recalculer le centre avec la moyenne des coordonnées de tous les points dans ce cluster $\rightarrow$ **REDUCER**

k-means et MapReduce

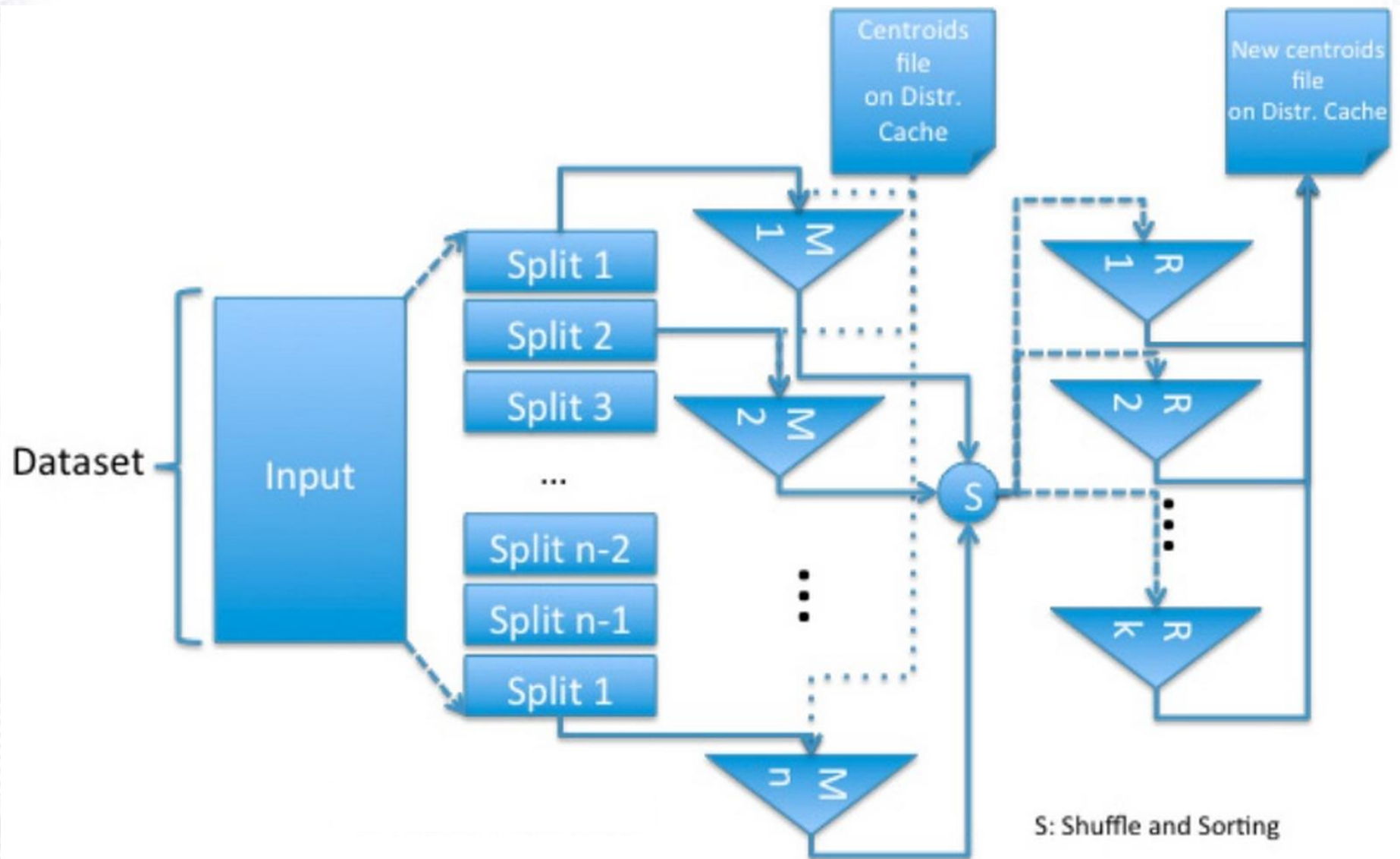
- Une seule itération de l'algorithme k-means:



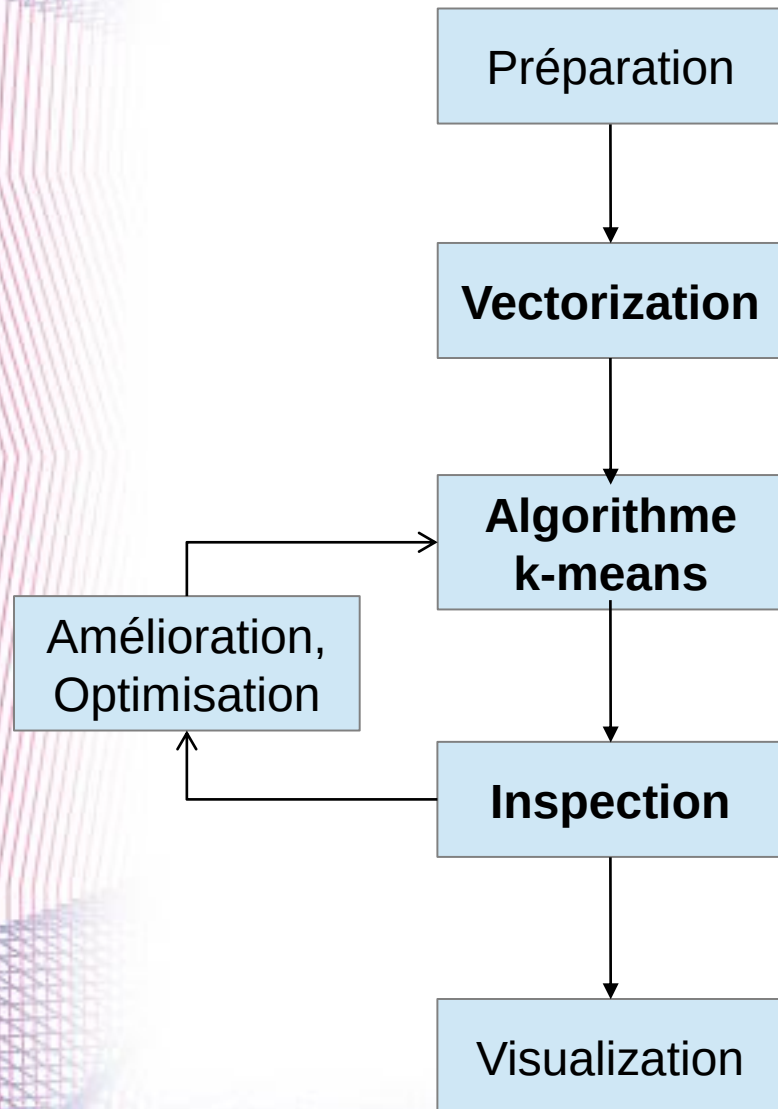
k-means et MapReduce

- La classe **KMeansDriver** itère sur les points et les clusters jusqu'à convergence des clusters (cluster ID codé « **Vn** ») ou nombre max d'itérations atteint
- La classe **KMeansMapper** affecte et output chaque point à son centre selon la mesure de distance spécifiée
output: (clusterID, ClusterObservation)
- La classe **KMeansCombiner** reçoit tous les paires (k,v) des mappers et effectue les **sommes partielles des vecteurs input** pour chaque cluster
output: (clusterID, ClusterObservation)
- La classe **KMeansReducer** reçoit les paires (k,v) des combiners, les somme afin de calculer le nouveau centre pour le cluster en question. Création du répertoire « cluster-X » à chaque itération
output: (Vn | Cn, Cluster)

k-means et MapReduce



k-means – Processus E2E

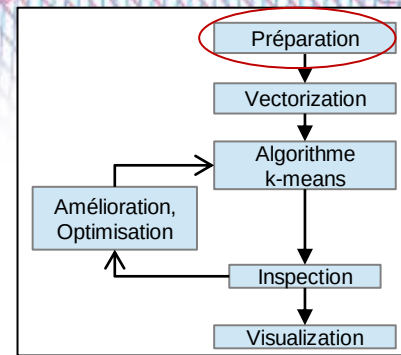


mahout seqdirectory e.g. conversion en SequenceFile
mahout seq2sparse e.g. vectorisation

mahout kmeans e.g. exécution + paramètres (k, condition d'arrêt, mesure de distance, etc.)

mahout clusterdump e.g. affichage des résultats

k-means Exemple



Téléchargement et extraction dans répertoire

AGENCE DE PRESSE REUTERS

<http://kdd.ics.uci.edu/databases/reuters21578/reuters21578.tar.gz>

22 fichiers SGML

Name	Size	Type
reuters-sgm	31 items	folder
reut2-000.sgm	1.3 MB	SGML document
reut2-001.sgm	1.3 MB	SGML document
reut2-002.sgm	1.2 MB	SGML document
reut2-003.sgm	1.3 MB	SGML document
reut2-004.sgm	1.3 MB	SGML document
reut2-005.sgm	1.4 MB	SGML document
reut2-006.sgm	1.3 MB	SGML document
reut2-007.sgm	1.3 MB	SGML document
reut2-008.sgm	1.4 MB	SGML document
reut2-009.sgm	1.3 MB	SGML document
reut2-010.sgm	1.4 MB	SGML document
reut2-011.sgm	1.3 MB	SGML document
reut2-012.sgm	1.3 MB	SGML document
reut2-013.sgm	1.1 MB	SGML document
reut2-014.sgm	1.1 MB	SGML document
reut2-015.sgm	1.3 MB	SGML document
reut2-016.sgm	1.3 MB	SGML document
reut2-017.sgm	1.5 MB	SGML document
reut2-018.sgm	1.3 MB	SGML document
reut2-019.sgm	1.3 MB	SGML document
reut2-020.sgm	1.0 MB	SGML document
reut2-021.sgm	621.6 kB	SGML document

1 000 documents

```

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u f BC-BAHIA-COCOA-REVIEW 02-26 0105</UNKNOWN>
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the Bahia cocoa zone, alleviating the drought since early
January and improving prospects for the coming temporaao,
although normal humidity levels have not been restored,
Comissaria Smith said in its weekly review.
...
Final figures for the period to February 28 are expected to
be published by the Brazilian Cocoa Trade Commission after
carnival which ends midday on February 27.
Reuter
&#3;</BODY></TEXT>
</REUTERS>
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&#5;&#5;&#5;F Y
&#22;&#22;&#1;f0708&#31;reute
d f BC-STANDARD-OIL-&lt;SRD>-TO 02-26 0082</UNKNOWN>
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<DATELINE> CLEVELAND, Feb 26 - </DATELINE><BODY>Standard Oil Co and BP North America
Inc said they plan to form a venture to manage the money market
borrowing and investment activities of both companies.
BP North America is a subsidiary of British Petroleum Co
Plc &lt;BP>, which also owns a 55 pct interest in Standard Oil.
The venture will be called BP/Standard Financial Trading
and will be operated by Standard Oil under the oversight of a
joint management committee.
...
  
```

k-means Exemple

Extraction des Reuters (documents textes)

```
$MAHOUT org.apache.lucene.benchmark.utils.ExtractReuters ${WORK_DIR}/reuters-smg ${WORK_DIR}/reuters-out
```



Name	Size	Type
reuters-smg	31 items	folder
reuters-out	21,578 items	folder
reut2-006.sgm-0.txt	680 bytes	plain text document
reut2-006.sgm-1.txt	651 bytes	plain text document
reut2-006.sgm-2.txt	376 bytes	plain text document
reut2-006.sgm-3.txt	607 bytes	plain text document
reut2-006.sgm-4.txt	251 bytes	plain text document
reut2-006.sgm-5.txt	688 bytes	plain text document
reut2-006.sgm-6.txt	777 bytes	plain text document
reut2-006.sgm-7.txt	502 bytes	plain text document
reut2-006.sgm-8.txt	357 bytes	plain text document
reut2-006.sgm-9.txt	259 bytes	plain text document
reut2-006.sgm-10.txt	338 bytes	plain text document
reut2-006.sgm-11.txt	182 bytes	plain text document
reut2-006.sgm-12.txt	94 bytes	plain text document
reut2-006.sgm-13.txt	447 bytes	plain text document
reut2-006.sgm-14.txt	608 bytes	plain text document
reut2-006.sgm-15.txt	1.0 kB	plain text document
reut2-006.sgm-16.txt	320 bytes	plain text document
reut2-006.sgm-17.txt	174 bytes	plain text document
reut2-006.sgm-18.txt	497 bytes	plain text document
reut2-006.sgm-19.txt	303 bytes	plain text document
reut2-006.sgm-20.txt	642 bytes	plain text document
reut2-006.sgm-21.txt	627 bytes	plain text document
reut2-006.sgm-22.txt	1.2 kB	plain text document
reut2-006.sgm-23.txt	1.2 kB	plain text document
reut2-006.sgm-24.txt	1.3 kB	plain text document
reut2-006.sgm-25.txt	96 bytes	plain text document
reut2-006.sgm-26.txt	476 bytes	plain text document
reut2-006.sgm-27.txt	857 bytes	plain text document
reut2-006.sgm-28.txt	194 bytes	plain text document
reut2-006.sgm-29.txt	422 bytes	plain text document

21 758 fichiers TEXTE

3-MAR-1987 09:18:21.26

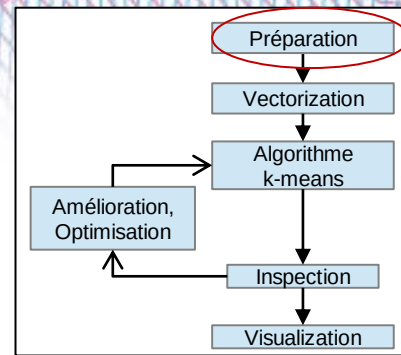
SANDOZ PLANS WEEDKILLER JOINT VENTURE IN USSR

Sandoz AG said it planned a joint venture to produce herbicides in the Soviet Union. The company said it had signed a letter of intent with the

3-MAR-1987 09:19:31.96

TAIWAN REJECTS TEXTILE MAKERS EXCHANGE RATE PLEA

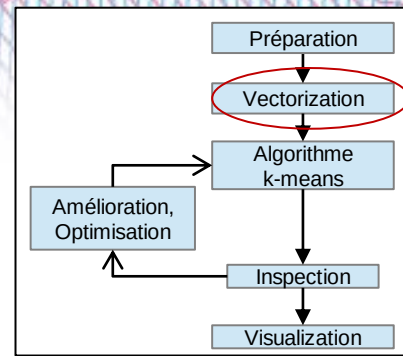
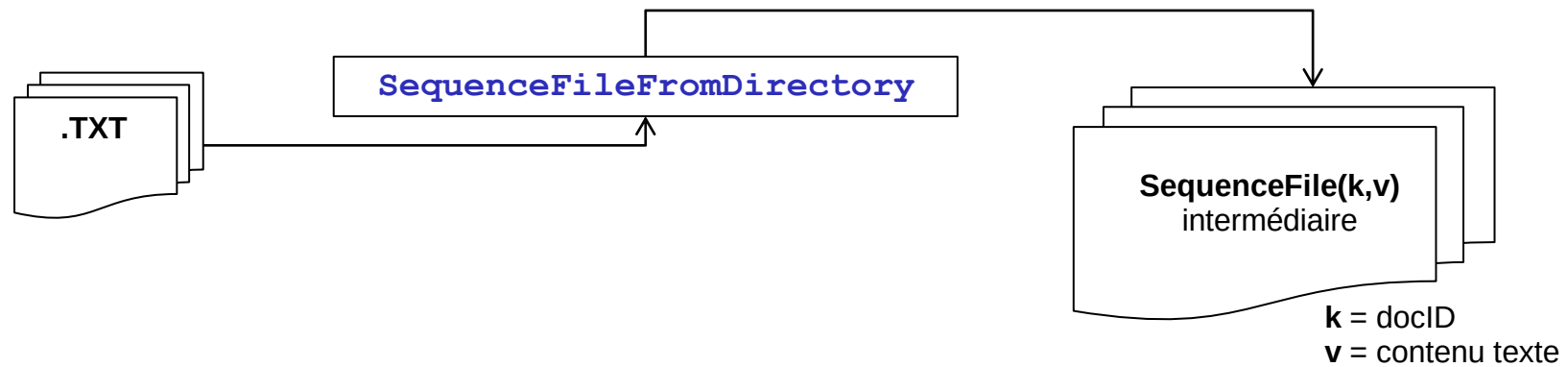
Central bank governor Chang Chi-cheng rejected a request by textile makers to halt the rise of the Taiwan dollar against the U.S. Dollar to stop them losing orders to South Korea, Hong Kong and Singapore, a spokesman for the Taiwan Textile Federation said. He quoted Chang as telling representatives of 19 textile associations last Saturday the government could not fix the Taiwan dollar exchange rate at 35 to one U.S. Dollar due to U.S. Pressure for an appreciation of the local currency. The Federation asked the government on February 19 to hold the exchange rate at that level. The federation said in its request that many local textile exporters were operating without profit and would go out of business if the rate continued to fall. Reuter



k-means Exemple

Conversion en SequenceFile avec seqdirectory

```
$MAHOUT seqdirectory -i ${WORK_DIR}/reuters-out -o ${WORK_DIR}/reuters-out-seqdir -c UTF-8 -chunk 64 -xm sequential
```



Name	Size	Type
▶ reuters-sgm	31 items	folder
▶ reuters-out	21,578 items	folder
▼ reuters-out-seqdir	2 items	folder
1 10 101 chunk-0	18.4 MB	unknown
1 10 101 .chunk-0.crc	143.8 kB	unknown

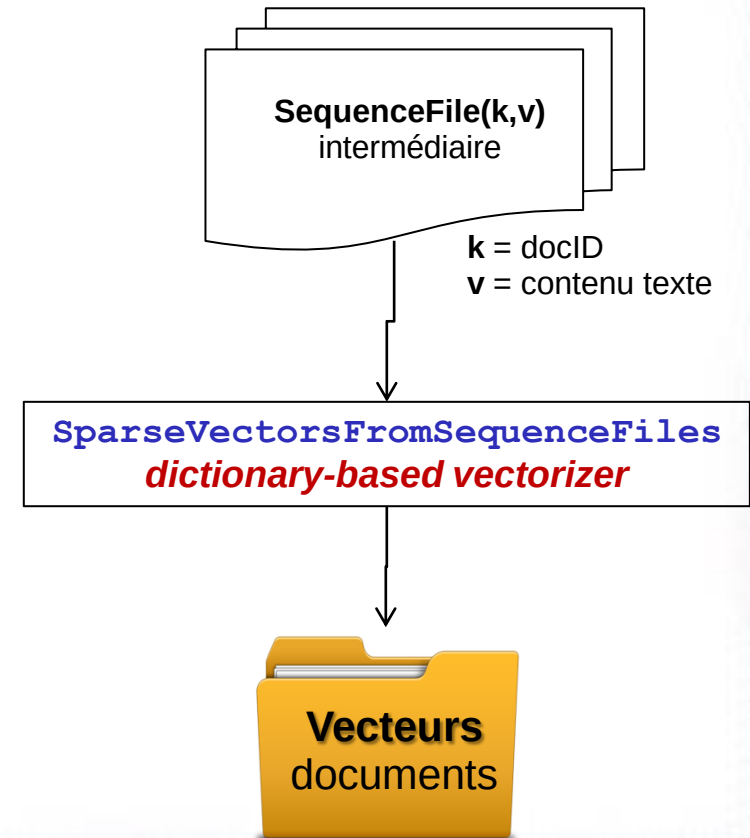
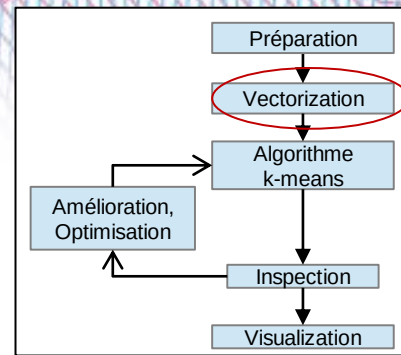
k-means Exemple

Vectorization avec seq2parse

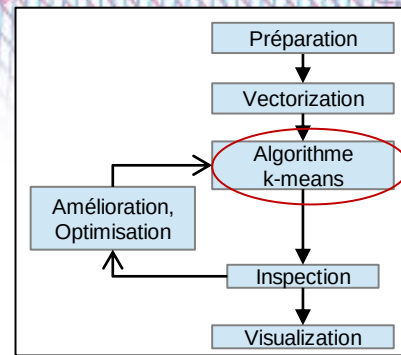
```
$MAHOUT seq2sparse \  
-i ${WORK_DIR}/reuters-out-seqdir/ \  
-o ${WORK_DIR}/reuters-out-seqdir-sparse-kmeans --maxDFPercent 85 --namedVector \  

```

Name	Size	Type
reuters-out	21,578 items	folder
reuters-out-seqdir	2 items	folder
reuters-out-seqdir-sparse-kmeans	9 items	folder
tokenized-documents	4 items	folder
part-m-00000	15.1 MB	unknown
_SUCCESS	0 bytes	plain text document
.part-m-00000.crc	117.8 kB	unknown
_SUCCESS.crc	8 bytes	unknown
wordcount	4 items	folder
part-r-00000	993.0 kB	unknown
_SUCCESS	0 bytes	plain text document
.part-r-00000.crc	7.8 kB	unknown
_SUCCESS.crc	8 bytes	unknown
df-count	4 items	folder
part-r-00000	844.6 kB	unknown
_SUCCESS	0 bytes	plain text document
.part-r-00000.crc	6.6 kB	unknown
_SUCCESS.crc	8 bytes	unknown
tf-vectors	4 items	folder
part-r-00000	17.0 MB	unknown
_SUCCESS	0 bytes	plain text document
.part-r-00000.crc	133.0 kB	unknown
_SUCCESS.crc	8 bytes	unknown
tfidf-vectors	4 items	folder
part-r-00000	17.0 MB	unknown
_SUCCESS	0 bytes	plain text document
.part-r-00000.crc	133.0 kB	unknown
_SUCCESS.crc	8 bytes	unknown
dictionary.file-0	824.1 kB	unknown
.dictionary.file-0.crc	6.4 kB	unknown
frequency.file-0	844.6 kB	unknown
.frequency.file-0.crc	6.6 kB	unknown



k-means Exemple



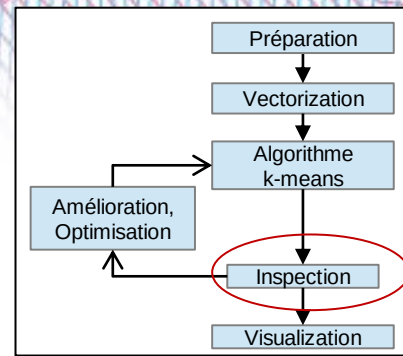
Exécution de l'algorithme k-means avec kmeans

```
$MAHOUT kmeans \
-i ${WORK_DIR}/reuters-out-seqdir-sparse-kmeans/tfidf-vectors/ \
-c ${WORK_DIR}/reuters-kmeans-clusters \
-o ${WORK_DIR}/reuters-kmeans \
-dm org.apache.mahout.common.distance.CosineDistanceMeasure \
-x 10 \
-k 20 \
-ow
--clustering \
```

- Centres (initiaux) générés avec RandomSeedGenerator
- Répertoires **clusters-*** créés à la fin de chaque itération; **informations sur les clusters**: centroid, écart-type, ...
- Répertoire **clusteredPoints** contient le mapping final **clusterID** $\leftrightarrow$ **docID**

reuters-kmeans-clusters		2 items	folder
part-randomSeed	25.3 kB	unknown	
.part-randomSeed.crc	208 bytes	unknown	
reuters-kmeans		7 items	folder
clusters-0		42 items	folder
clusters-1		6 items	folder
clusters-2-final		6 items	folder
part-r-00000	3.6 MB	unknown	
._policy	194 bytes	unknown	
._SUCCESS	0 bytes	plain text document	
.part-r-00000.crc	27.8 kB	unknown	
._policy.crc	12 bytes	unknown	
._SUCCESS.crc	8 bytes	unknown	
clusteredPoints		4 items	folder
part-m-00000	16.8 MB	unknown	
._SUCCESS	0 bytes	plain text document	
.part-m-00000.crc	131.2 kB	unknown	
._SUCCESS.crc	8 bytes	unknown	
._policy	194 bytes	unknown	
._policy.crc	12 bytes	unknown	
clusterdump	29.3 kB	plain text document	

k-means Exemple



Visualisation avec clusterdump

```

$MAHOUT clusterdump \
-i ${WORK_DIR}/reuters-kmeans/clusters-*-final \
-o ${WORK_DIR}/reuters-kmeans/clusterdump \
-d ${WORK_DIR}/reuters-out-seqdir-sparse-kmeans/dictionary.file-0 \
-dt sequencefile -b 100 -n 20 --evaluate -dm org.apache.mahout.common.distance.CosineDistanceMeasure -sp 0 \
--pointsDir ${WORK_DIR}/reuters-kmeans/clusteredPoints \

```

```

:VL-7959{n=1601 c=[0.1:0.004, 0.1365:0.012, 0.18:0.006, 0.4:0.004, 0.523:0.012, 0.55:0.006, 0.8:0.004

```

Top Terms:

shares	=>	3.0701310359113743
common	=>	2.055063796147639
stock	=>	1.9897432817212497
said	=>	1.9861087525062155
company	=>	1.5980280315332454
inc	=>	1.5619777712801113
its	=>	1.5518101159667015
dlrs	=>	1.4305196776381142
group	=>	1.394632834482759
min	=>	1.3522909689813312
pct	=>	1.3290586551973627
stake	=>	1.3190966131983513
share	=>	1.2258259995440852
has	=>	1.1129996320890085
corp	=>	1.090765117482049
buy	=>	1.0501346308167914
outstanding	=>	1.0468410090458982
securities	=>	0.946948390986903
exchange	=>	0.9295935481879802
mar	=>	0.8751847061941134

Weight : [props - optional]: Point:

```

:VL-16919{n=683 c=[0.1:0.010, 0.10:0.022, 0.15:0.012, 0.18:0.013, 0.2:0.010, 0.20:0.013, 0.24:0.014,

```

Top Terms:

pct	=>	3.36537534495922
1	=>	3.004444594264554
bond	=>	2.89077234617264
8	=>	2.573712494153404
manager	=>	2.5478165704718796
lead	=>	2.5447139586397007
coupon	=>	2.5384799090171803

...

```

:VL-21446{n=1026 c=[0.2:0.006, 0.25:0.007, 0.5:0.031, 0.0:0.059, 0.05:0.009, 0.07:0.009, 0.12:0.010

```

Top Terms:

said	=>	1.89790412801051
inc	=>	1.4305799771935386
company	=>	1.3819211493691042
its	=>	1.29199122685438
sales	=>	1.2823967536290486
dlrs	=>	1.2325192882768359
Drug	=>	1.1648731208451775

..

Inter-Cluster Density: 0.640986736268577

Intra-Cluster Density: 0.7052966012163074

CDbw Inter-Cluster Density: 0.0

CDbw Intra-Cluster Density: 18.296697402501465

CDbw Separation: 235.31938537237372

reuters-kmeans-clusters		2 items	folder
part-randomSeed	25.3 kB	unknown	
.part-randomSeed.crc	208 bytes	unknown	
reuters-kmeans		7 items	folder
clusters-0	42 items	folder	
clusters-1	6 items	folder	
clusters-2-final	6 items	folder	
part-r-00000	3.6 MB	unknown	
_policy	194 bytes	unknown	
_SUCCESS	0 bytes	plain text document	
.part-r-00000.crc	27.8 kB	unknown	
._policy.crc	12 bytes	unknown	
._SUCCESS.crc	8 bytes	unknown	
clusteredPoints	4 items	folder	
part-m-00000	16.8 MB	unknown	
_SUCCESS	0 bytes	plain text document	
.part-m-00000.crc	131.2 kB	unknown	
._SUCCESS.crc	8 bytes	unknown	
._policy	194 bytes	unknown	
._policy.crc	12 bytes	unknown	
clusterdump	29.3 kB	plain text document	

k-means Exemple

Visualisation avec `clusterdump`

Listing 10.1. Top 10 terms using k-means clustering with a **Euclidean distance measure**

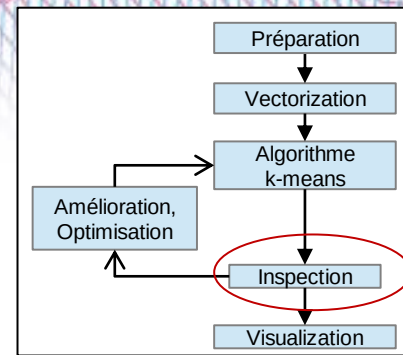
```
Top Terms:
said          => 11.60126582278481
bank          => 5.943037974683544
dollar        => 4.89873417721519
market        => 4.405063291139241
us            => 4.2594936708860756
banks         => 3.3164556962025316
pct           => 3.069620253164557
he            => 2.740506329113924
rates         => 2.7151898734177213
rate          => 2.7025316455696204
```

The top terms in the preceding cluster indicate that it contains documents related to banks. A similar cluster is created using a cosine distance measure in the following listing.

Listing 10.2. Top 10 terms using k-means clustering and a **cosine distance measure**

```
Top Terms:
bank          => 3.3784878432986294
stg           => 3.122450990639185
bills         => 2.440446103514419
money         => 2.324820905806048
market        => 2.223828649332401
england       => 1.6710182027854468
pct           => 1.5883359918481277
us            => 1.490838685054553
dealers       => 1.4752549691633745
billion       => 1.3586127823991738
```

They're similar, but the second cluster appears better than the first one. The second cluster identifies *money* as an important term, whereas the first cluster has a common verb *said* as the top term.



K-means Exemple

- **Ubuntu 12.04 LTS**
- **Mahout 0.8**
- **Java 1.7_0.25**
- **Cloudera Hadoop 2.0.0 CDH4.0.5**
- **Eclipse “Kepler” Service Release 1**
 - **m2e – Maven integration for Eclipse**

Next Steps

- **Exécuter sur Hadoop**

- local e.g. MAHOUT_LOCAL=TRUE
- configuration *pseudo-distributed*
- cluster de machines



not there yet ...

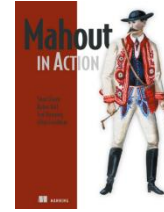
- **Qualité de données**

- Tester avec plusieurs mesures de distance
- Tester avec différentes stratégies de pré-traitements e.g. n-gram, normalisation, etc.

Bibliographie

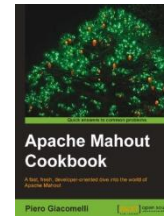
- **Mahout In Action**

By: **Sean Owen; Robin Anil; Ted Dunning; Ellen Friedman**
Publisher: Manning Publications
Pub. Date: October 14, **2011**
Print ISBN-13: 978-1-935182-68-9



- **Apache Mahout Cookbook**

By: **Piero Giacomelli**
Publisher: Packt Publishing
Pub. Date: December 26, **2013**
Print ISBN-13: 978-1-84951-802-4



- **Parallel K-Means Clustering Based on MapReduce**

Weizhong Zhao, Huifang Ma, and Qing He
CloudCom **2009**, LNCS 5931, pp. 674–679
Springer-Verlag Berlin Heidelberg

- **MapReduce Design of K-Means Clustering Algorithm**

Prajesh P Anchalia, Anjan K Koundinya, Srinath N K
R V College of Engineering, Bangalore, India
IEEE 2013

Merci ! ... Q&A

Backup slides

k-means - Vectorization

- Mahout implémente 3 classes de vecteurs:
 - `DenseVector` can be thought of as an array of doubles, whose size is the number of features in the data. Because all the entries in the array are preallocated regardless of whether the value is 0 or not, we call it *dense*.
 - `RandomAccessSparseVector` is implemented as a `HashMap` between an `integer` and a `double`, where only nonzero valued features are allocated. Hence, they're called as `SparseVector` s.
 - `SequentialAccessSparseVector` is implemented as two parallel arrays, one of integers and the other of doubles. Only nonzero valued entries are kept in it. Unlike the `RandomAccessSparseVector`, which is optimized for random access, this one is optimized for linear reading.

Utilisé pour k-means

Introduction à l'algo k-means

Remarques:

- **Complexité importante**
 - $k \times n \times O(\text{distance}) \times \text{nb d'itérations}$
- **La solution finale dépend des représentants initiaux choisis;**
un choix aléatoire rend l'algorithme non déterministe
 - On exécute plusieurs fois et on choisit la partition qui minimise la distance intraclasse
- **Pas de minimum global** de la distance intra-classe, seul le minimum local est atteint.
 - Difficile d'imposer une partition stable à cause des problèmes d'arrondi informatique, alors on applique la condition sur le nb max d'itérations

Apache Mahout k-means

- Mesures de distance supportées:
 - *Euclidean distance measure*
 - *Squared Euclidean distance measure*
 - *Manhattan distance measure*
 - *Cosine distance measure*
 - *Tanimoto distance measure*
 - *Weighted distance measure*
- *Le type de distance est un facteur déterminant dans la qualité des clusters*